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Markets Overview

	Week Ago	Today
DOW	35,209	35,314
S&P	4,437	4,445
NASDAQ	14,836	14,694
RUT	2,248	2,195

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	Week Ago	Today
AMZN	3,345	3,250
AAPL	146	149
GOOG	2,741	2,757
MSFT	289	293
FEB	364	363
NFLX	521	514
ZM	383	348
DAL	40.32	39.36

(3)

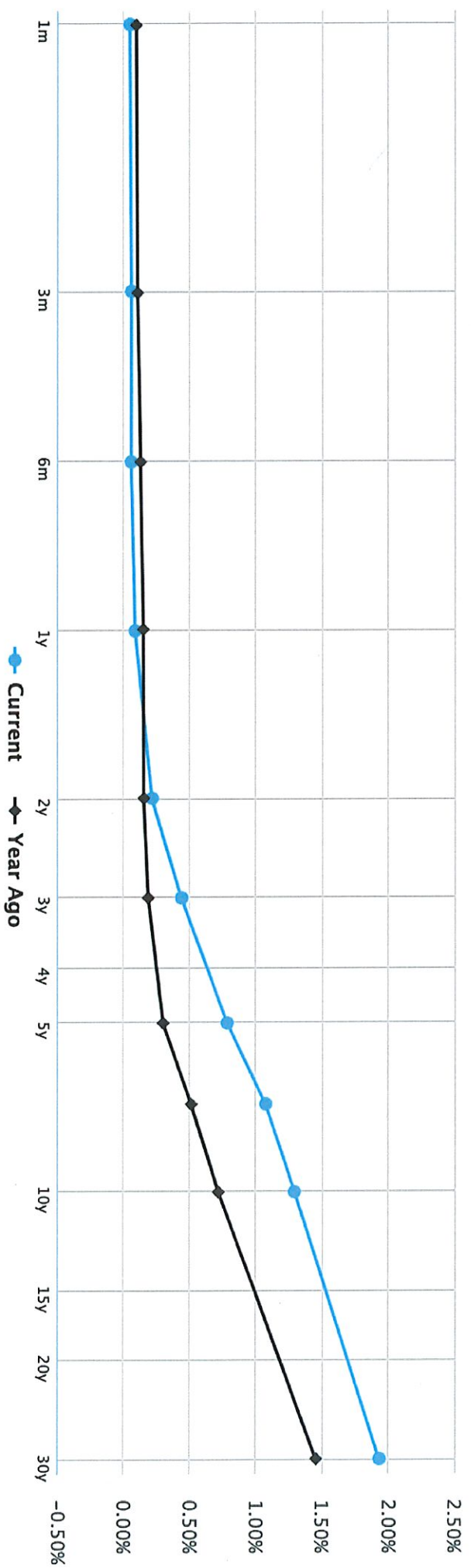
	Week Ago	Today
TSLA	699	685 ⁺
XOM	57.86	55.72
JNJ	173	176.98
PFE	45.07	47.85
MRNA	414	354
GS	398	407
C	71	72

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US Treasury Rates

Maturity	Week Ago	Today
3m	.06%	.05%
6m	.06%	.05%
1 yr	.09%	.08%
2 yr	.21%	.20%
5 yr	.77%	.74%
10 yr	1.31%	1.25%
20 yr	1.85%	1.81%
30 yr	1.94%	1.90%

Yield Curve



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Currencies

	Last Week	This Week
EUR	1 EUR = 1.175 USD	1.178
JPY	1 USD = 110.24 JPY	109.19
GBP	1 GBP = 1.385 USD	1.384
CHF	1 USD = .919 CHF	.912
CAD	1 USD = 1.257 CAD	1.257
CNY	1 USD = 6.485 CNY	6.476
BTC	1 BTC = 45,589 USD	46,400

Commodities

	Last Week	This Week
WTI Crude	65.61	66.71
Brent Crude	68.19	69.02
Gold	1,732	1,786
Lumber	530.10	495.20

52 Week High 1,670.50

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VIX

Implied Volatility on 30 Day
S&P Options

Last Week 17.10%

Today 17.15%

All Time High 89.53%

All Time Low 8.56%

Typical $\approx 15\% - 20\%$

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Historical Volatility

Pick a time window (ending now, usually)

Daily Returns $R_n = \frac{S_n - S_{n-1}}{S_{n-1}}$

S_n Stock Price on Day n

Notice $1 + R_n = \frac{S_n}{S_{n-1}}$

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$$\text{Let } \chi_n = \ln(1+R_n) = \ln\left(\frac{S_n}{S_{n-1}}\right)$$

$$\boxed{\ln(1+R_n) \approx R_n \quad \ln(1+x) \approx x \quad x \text{ small}}$$

$$\text{Let } \hat{\mu}_d = \frac{1}{N} \sum_{n=1}^N \chi_n \quad \text{Sample Mean}$$

$$\hat{\sigma}_d^2 = \frac{1}{N-1} \sum_{n=1}^N [\chi_n - \hat{\mu}_d]^2$$

← annualized

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$$\sigma_H = \sqrt{252} \quad \hat{\sigma}_d = \sqrt{252} \text{ Std Dev } \ln\left(\frac{S_n}{S_{n-1}}\right)$$

historical
volatility

252 Trading Days

Mathematicians Trading
Year 256 day

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Yields & Bonds

Present Time $t=0$, $T > 0$

$d(T)$ = price at time 0 to receive
\$1 at T (no risk of
default)

Discount Factor for Time T

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Zero Coupon Bond Face value F

Maturity T , receive F at T

Time-0 Price $F_d(T)$

Coupon Bonds T -multiple of
6 months

Maturity T , Face F
annualized coupon rate q

Get $F \frac{q}{2}$ every 6 months

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Also get F at time T

Time-0 Price

$$F d(T) + F \frac{q}{2} \sum_{i=1}^{2T} d(\frac{i}{2})$$

T fixed and $q \uparrow$ then Price $\uparrow$

Usually q chosen to make

$P \approx F$ when bond is issued

q called Coupon yield

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Most Recently Issued of
each maturity "on the run"
Older Bonds "off the run"

$P > F$ Premium

$P = F$ Par

$P < F$ Discount

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Treasury Strips

STRIPS - Separate Trading
of Registered Interest and
Principal Securities

C-Strips Coupons

P-Strips Principal

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C-Strips, P-Strips for same maturity and same amount often trade at different prices.

Can Reassemble Strips to make a bond

Need Exact P-Strip (Same Cusip #)

Coupon Strips are Interchangeable

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Semiannual Compounding

$(1 + \frac{r}{2})^{2T}$ = value of \$1
after T years

Different Rates for Different
Maturities T

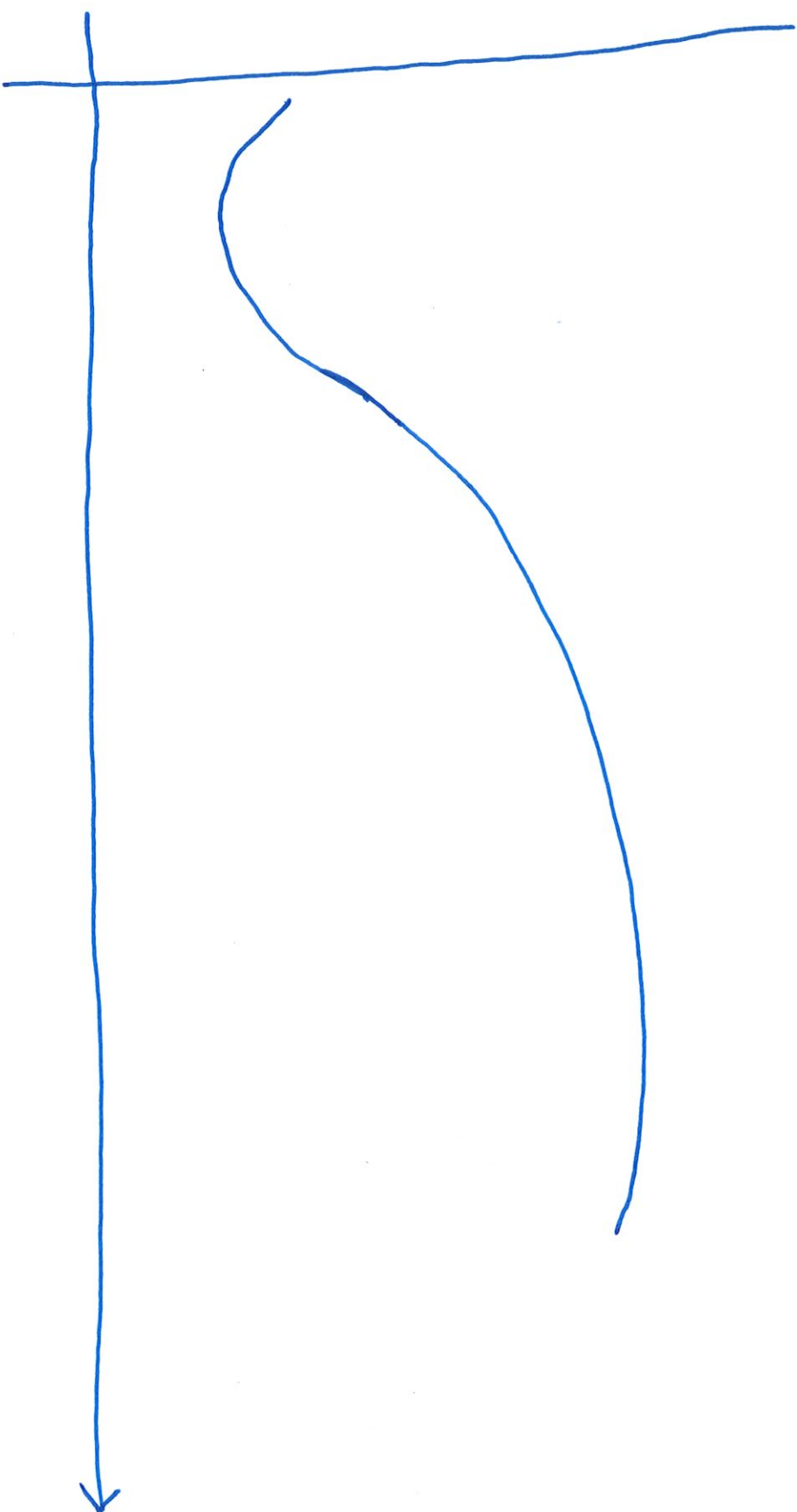
Spot Rate for Maturity T
(single payment at T)

$\hat{r}(T)$
Spot rate
for T

$$d(T) = \frac{1}{(1 + \frac{\hat{r}(T)}{2})^{2T}}$$

Spot Rate Curve from 2001

$18\frac{1}{2}$



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Yield to Maturity y

$$P = \frac{F}{(1+\frac{y}{2})^{2T}} + F \frac{1}{2} \sum_{i=1}^{2T} \frac{1}{(1+\frac{y}{2})^i}$$

market
Price $\rightarrow$

Coupon Bond

y -yield to maturity

"average" spot rate

Magic value of y s.t. if $\hat{r}(t) = y$
for all t then Market Price
= Computed price

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$$\begin{aligned} & \cancel{F} d(T) + \cancel{F} \frac{q}{2} \sum_{i=1}^{2T} d\left(\frac{i}{2}\right) \\ &= \frac{\cancel{F}}{\left(1 + \frac{y}{2}\right)^{2T}} + \cancel{F} \frac{q}{2} \sum_{i=1}^{2T} \frac{1}{\left(1 + \frac{y}{2}\right)^i} \end{aligned}$$

Face F cancels, but q
does not!

y depends on q and T

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Thm:

$$(i) P > F$$



$$q > y$$

$$(ii) P = F$$



$$q = y$$

$$(iii) P < F$$



$$q < y$$

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Compute Par-Coupon Yield

$$\cancel{F} = \cancel{F}d(T) + \cancel{F} \frac{q}{2} \sum_{i=1}^{2T} d(\frac{i}{2})$$

Solve for q

$$q = \frac{2[1-d(T)]}{\sum_{i=1}^{2T} d(\frac{i}{2})}$$

By Thm

$$q = y$$

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Par-Coupon Yield
= Plain Vanilla
Swap Rate

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DV01 "Dollar Value of
an '01 (basis point)

Assume Bond with Fixed Payments

Changes in yields measured
in bp

100 bp = 1 Percentage Point

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$$DVO1 = \frac{-\text{Change in Price}}{\text{Change in yield in bp}}$$

yield $\nearrow$ Price $\searrow$

Rule of Thumb

10 yr Treasury typically
has DVO1 \$8 to \$9
per \$10,000 face
8¢ to 9¢ per \$100 face

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Today \$9.37 per 10,000 face

EX: Hold 1,000,000 F of 10 yr
y drops 3 bp.

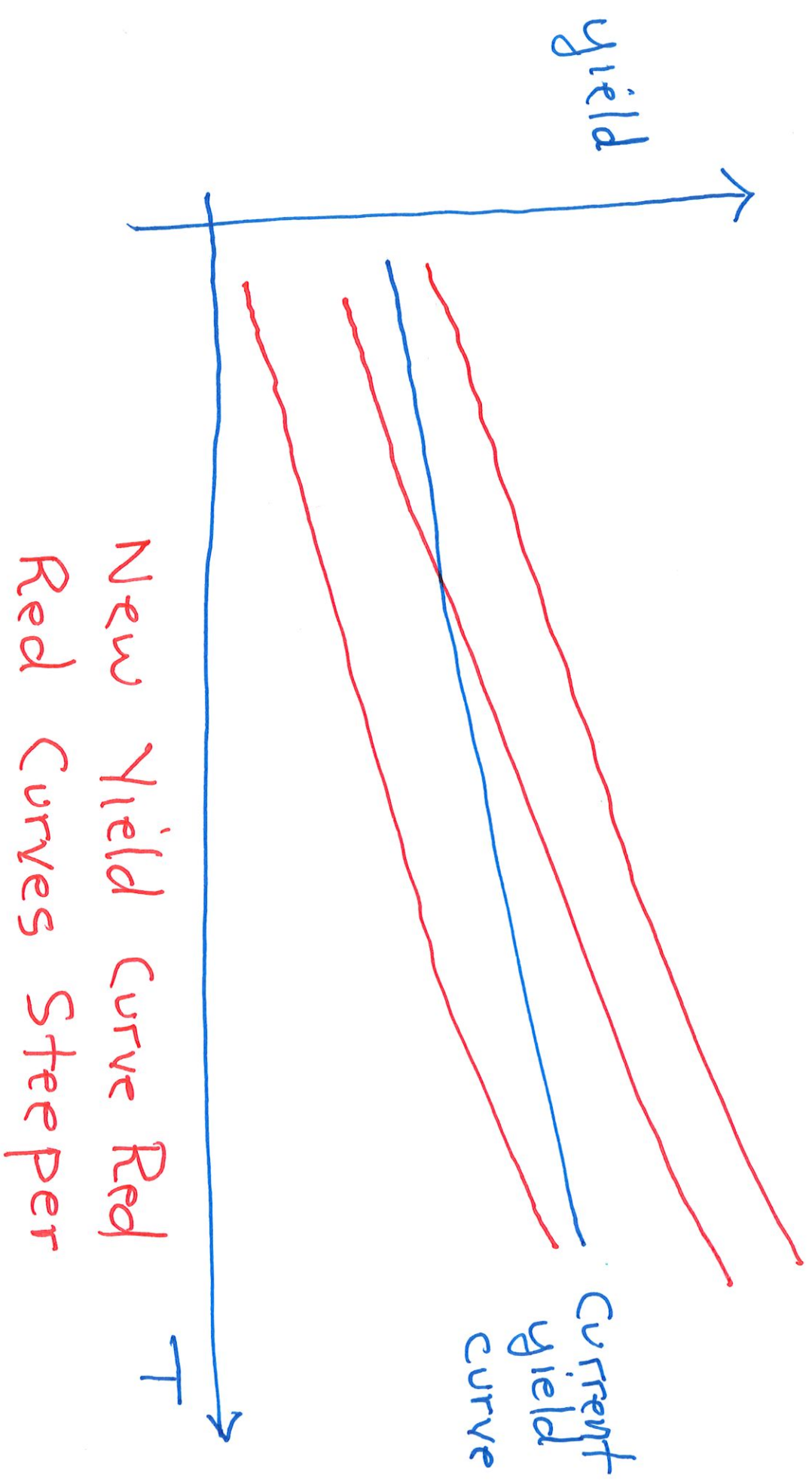
$$\text{Make} \approx 3(9.37)100 = 2,811$$

With $y = 2\%$ $DV01 = 9.022$
per 10,000 F

With $y = 4\%$ $DV01 = 8.175$
per 10,000 F

10 yr Treasury

Look at Steepeners
and Flatteners



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Steeper $T_1 < T_2$

(Believe $y(T_2)$ increase relative to $y(T_1)$)

Buy T_1 Mat Bond

Sell T_2 Mat Bond

DY01's 2 Bonds Same

DY01 Portfolio is Zero

Dvol T_1 -yr bond

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$$\text{Appx Profit} = (Dvol) \cdot \Delta$$

Δ = # bps Steepening

change $y(T_2)$ - change $y(T_1)$
measured in bp.

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Flatteners

Sell T_1 -mat bond

Buy T_2 -mat bond

Match DY01s

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Can Implement Steepeners
& Flatteners with
Cash Treasuries or
Futures

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Butterflies

$$T_1 < T_2 < T_3$$

Position on T_2 yr bond

Opposite Positions on T_1, T_3 yr bonds

Needs 2 Conditions to Choose Weights

(i) Zero DVOI for Portfolio

(ii) Several Popular Choices

(Could use cash neutral, zero cost)

Futures

Underlying Asset
Maturity (Measure time in days)

Idea: Lock in a price now to
buy or sell asset on
day N (maturity)

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Day n Futures Price $F_{n,N}$

Long Position Buy

Short Position Sell

Cost to enter is 0

but you must post margin
(or collateral)

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At end of day n

Long Position Receives

$$Fut_{n,N} - Fut_{n-1,N}$$

$$Fut_{N,N} = \text{Asset Price on Day } N \\ (\text{Spot Price})$$

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Neglect Interest on Margin Acct

$$\cancel{F_{ut_{1,N}}} - F_{ut_{0,N}} + (\cancel{F_{ut_{2,N}}} - \cancel{F_{ut_{1,N}}}) + \dots$$

$$\dots + \cancel{F_{ut_{n,N}}} - \cancel{F_{n-1,N}}$$

$$\dots + \cancel{F_{ut_{N,N}}} - \cancel{F_{ut_{N-1,N}}}$$

$$= F_{ut_{N,N}} - F_{ut_{0,N}}$$

$$= \text{Spot Price} - F_{ut_{0,N}}$$

Treasury Futures

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ZT	2 yr	(1 $\frac{3}{4}$ yrs to 2 yrs)
Z3N	3 yr	(2 $\frac{3}{4}$ yrs to 3 yrs)
ZF	5 yr	(4 yrs 3m to 5 yrs 3m)
ZN	10 yr	(6.6 yrs to 10 yrs)
TN	Ultra10	(9.5 yrs to 10 yrs)
ZB	Bond	(15 yrs to 25 yrs)
UB	Ultrabond	(25 yrs to 30 yrs)

~ Based on Bonds with
hypothetical 6% Coupons

Z_T, Z_{3N} Each Contract
is \$200,000 F

For Z_F, Z_N, T_N, Z_B, U_B Each
Contract is for \$100,000
face.